Prob. 14.4

Sodium hydride NaH structure.

Na atoms form FCC lattice

H atoms displaced from Na by $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ (ZnS structure)

or by $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ (NaCl structure)

ZnS is described in example 4, page 149.

NaCl is described in page 124.

Solve by using Eq. (14.14) on page 149:

$$S_{hkl} = S^{Lattice}_{hkl} \cdot S^{basis}_{hkl}$$

Thus for NaCl-type structure $S^{basis}_{hkl} = f_{Na} +$

$$+ f_H \, e^{i 2\pi (h,k,l) \cdot (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})} =$$

$$= f_{Na} + f_H \, e^{i\pi(h+k+l)} = S^{basis}_{NaCl}$$

For ZnS-type structure $S^{basis}_{hkl} = f_{Na} + f_H \, e^{i 2\pi (h,k,l)(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})}$

$$= f_{Na} + f_H \, e^{i \frac{\pi}{2}(h+k+l)} = S^{basis}_{ZnS}$$

For <u>both</u> of them, $S^{Lattice}_{hkl} = 1 + e^{i\pi(h+k)} + e^{i\pi(h+l)} + e^{i\pi(k+l)}$

is the same, see eq. (14.13) in the book.

Therefore for intensities $I_{hkl} = |S_{hkl}|^2 =$

$= |S^{Lattice}_{hkl}|^2 \, |S^{basis}_{hkl}|^2$. Since we are interested

in the ratio of two intensities, $S^{Lattice}_{hkl}$ cancels out.

Therefore, intensity ratio $\frac{I_{(111)}}{I_{(200)}} = \frac{|S^{basis}_{111}|^2}{|S^{basis}_{200}|^2}$

But $S^{basis}_{ZnS} = f_{Na} + f_H e^{i\frac{\pi}{2}(1+1+1)} = f_{Na} - i f_H$ for (111),

$\Rightarrow |S^{basis}_{ZnS}|^2 = f_{Na}^2 + f_H^2$ for $(hkl) = (111)$

For $(hkl) = (200)$ $S^{basis}_{ZnS} = f_{Na} + f_H e^{i\frac{\pi}{2}\cdot 2} =$

$= f_{Na} - f_H$

$\Rightarrow$ for ZnS structure $\frac{I_{111}}{I_{200}} = \frac{f_{Na}^2 + f_H^2}{(f_{Na} - f_H)^2} =$

$= \frac{0.363^2 + 0.374^2}{(0.363 - (-0.374))^2} \cong \frac{0.27}{0.54} = 0.5$

For NaCl structure we obtain:

$S^{basis}_{NaCl}(hkl = 111) = f_{Na} + f_H e^{i\pi(1+1+1)} = f_{Na} - f_H$

$S^{basis}_{NaCl}(hkl = 200) = f_{Na} + f_H e^{i\pi(2)} = f_{Na} + f_H$

$\Rightarrow \frac{I_{111}}{I_{200}} = \frac{(f_{Na} - f_H)^2}{(f_{Na} + f_H)^2} = \frac{(0.363 - (-0.374))^2}{(0.363 - 0.374)^2} = \frac{0.54}{0.00012}$

≈ 4462

Therefore here $I_{111} \gg I_{200} \Rightarrow$ NaH forms NaCl structure

Powder diffraction is also sensitive to multiplicity factors: $(hkl) = (111)$ is the same

as $(\bar{1}11)$, $(1\bar{1}1)$, $(11\bar{1})$, $(\bar{1}\bar{1}1)$, $(\bar{1}1\bar{1})$
$(1,\bar{1},\bar{1})$, $(\bar{1},1\bar{1})$

$\Rightarrow$ multiplicity $= 8$.

For $(hkl) = (200)$ we need to account for
(020), (002), $(\bar{2}00)$, $(0\bar{2}0)$, $(00\bar{2})$

$\Rightarrow$ multiplicity $= 6$.

But $\frac{8}{6} = \frac{4}{3} = 1.3$ is not a large factor
and will not change the conclusion that

$$\frac{I_{111}}{I_{200}} \text{ for NaCl-structure} \gg \frac{I_{111}}{I_{200}} \text{ for ZnS struct.}$$

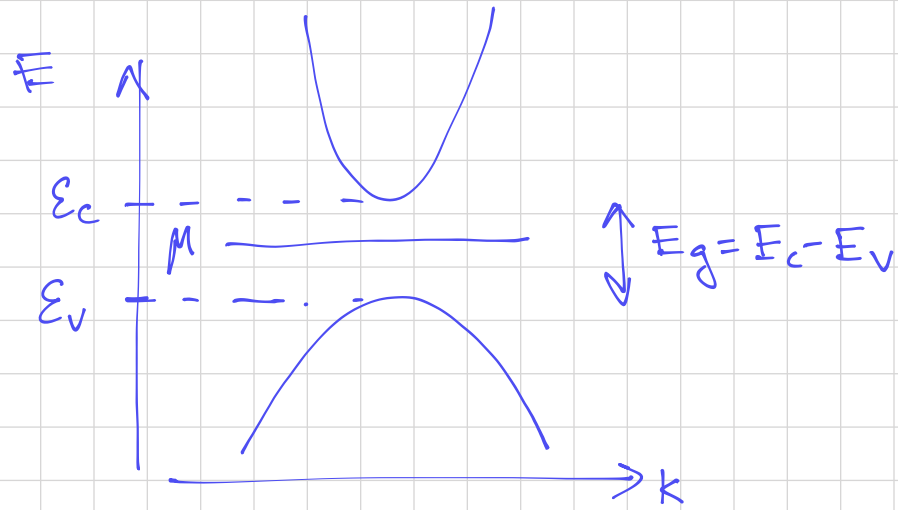
Prob. 17.2

(a) this problem follows closely calculations in <u>17.3</u>

For parabolic band with $\varepsilon_k = \frac{\hbar^2 k^2}{2m}$

$$g_c(\varepsilon) = \frac{(2m_e)^{3/2}}{2\pi^2\hbar^3}\sqrt{\varepsilon - \varepsilon_c}$$

$$g_v(\varepsilon) = \frac{(2m_h)^{3/2}}{2\pi^2\hbar^3}\sqrt{\varepsilon_v - \varepsilon}$$



Density of electrons $n(T) = \int_{\varepsilon_c}^{\infty} d\varepsilon \, \frac{g_c(\varepsilon)}{e^{\beta(\varepsilon-\mu)}+1} \approx \int_{\varepsilon_c}^{\infty} d\varepsilon \, g_c(\varepsilon)\, e^{-\beta(\varepsilon-\mu)}$ for $\beta(\varepsilon_c - \mu) \gg 1$;

holes $$p(T) = \int_{-\infty}^{\varepsilon_v} d\varepsilon \, g_v(\varepsilon)\left(1 - \frac{1}{e^{\beta(\varepsilon-\mu)}+1}\right) =$$

$$= \int_{-\infty}^{\varepsilon_v} d\varepsilon \, g_v(\varepsilon) \frac{e^{\beta(\varepsilon-\mu)}}{e^{\beta(\varepsilon-\mu)}+1} \approx \int_{-\infty}^{\varepsilon_v} d\varepsilon \, g_v(\varepsilon)\, e^{\beta(\varepsilon-\mu)}$$

for $\mu > \varepsilon_v$ and $e^{\beta(\mu-\varepsilon_v)} \gg 1$.

Then for $\begin{cases} n(T) & \text{introduce } x = \varepsilon - \varepsilon_c \\ p(T) & \longrightarrow y = \varepsilon_v - \varepsilon \end{cases}$

So $$n(T) = \frac{(2m_e)^{3/2}}{2\pi^2\hbar^3}\int_0^{\infty} dx \sqrt{x}\, e^{-\beta(x+\varepsilon_c-\mu)} = (x = y^2)$$

$$= \frac{(2m_e)^{3/2}}{\pi^2\hbar^3} e^{-\beta(\varepsilon_c-\mu)} \underbrace{\int_0^{\infty} dy\, y^2 e^{-\beta y^2}}_{J(\beta)} = \frac{(2m_e k_B T)^{3/2}}{4\pi^{3/2}\hbar^3} e^{-\beta(\varepsilon_c-\mu)}$$

Observe that $J(\beta) = -\frac{\partial}{\partial\beta}\int_0^{\infty} dy\, e^{-\beta y^2} = -\frac{\sqrt{\pi}}{2}\frac{\partial}{\partial\beta}\beta^{-1/2} = \frac{\sqrt{\pi}}{4}\beta^{-3/2}$

But $\int_0^{\infty} dy\, e^{-\beta y^2} = \frac{\sqrt{\pi}}{2}\beta^{-1/2}$

Similarly $p(T) = \int_{-\infty}^{\varepsilon_v} d\varepsilon \; g_v(\varepsilon)\, e^{-\beta(\mu-(\varepsilon-\varepsilon_v)-\varepsilon_v)} = (y = \varepsilon_v - \varepsilon)$

$$= -\int_{\infty}^{0} dy \; g_v(\varepsilon_v - y)\, e^{-\beta y}\, e^{-\beta(\mu-\varepsilon_v)} = \frac{(2m_h)^{3/2}}{2\pi^2\hbar^3}\, e^{-\beta(\mu-\varepsilon_v)} \int_0^{\infty} dy \sqrt{y}\, e^{-\beta y}$$

but $$g_v(\varepsilon) = \frac{(2m_h)^{3/2}}{2\pi^2\hbar^3}\sqrt{\varepsilon_v - \varepsilon} = \frac{(2m_h)^{3/2}}{2\pi^2\hbar^3}\sqrt{y}$$

$$\Rightarrow p(T) = \frac{(2m_h k_B T)^{3/2}}{4\pi^{3/2}\hbar^3}\, e^{-\beta(\mu-\varepsilon_v)}$$

So that $$n(T)\,p(T) = \frac{(2m_e k_B T)^{3/2}}{4\pi^2\hbar^3}\,\pi\,\frac{(2m_h k_B T)^{3/2}}{4\pi^2\hbar^3}\, e^{-\beta(\varepsilon_c-\mu)} \cdot e^{-\beta(\mu-\varepsilon_v)}$$

$$= \frac{(2k_B T)^3}{16\pi^3\hbar^6}\,(m_e m_h)^{3/2}\, e^{-\beta(\varepsilon_c-\varepsilon_v)}$$

$$= \frac{1}{2}\left(\frac{k_B T}{\pi\hbar^2}\right)^3 (m_e m_h)^{3/2}\, e^{-\beta E_g}, \quad E_g = \varepsilon_c - \varepsilon_v \text{ gap}$$

In the intrinsic semiconductor $n(T) = p(T)$

$$\Rightarrow n(T) = \frac{1}{\sqrt{2}}\left(\frac{k_B T}{\pi\hbar^2}\right)^{3/2}(m_e m_h)^{3/4}\, e^{-\frac{E_g}{2k_B T}}.$$

We can also take the ratio: $$1 = \frac{n(T)}{p(T)} = \left(\frac{m_e}{m_h}\right)^{3/2} \frac{e^{-\beta(\varepsilon_c-\mu)}}{e^{-\beta(\mu-\varepsilon_v)}}$$

$$\Rightarrow e^{\beta(2\mu-\varepsilon_c-\varepsilon_v)} = \left(\frac{m_h}{m_e}\right)^{3/2}$$

$$\Rightarrow \mu = \frac{1}{2}(\varepsilon_c + \varepsilon_v) + \frac{3}{4}k_B T \ln\left(\frac{m_h}{m_e}\right)$$

At $T=0$ the chemical potential is in the middle of the gap separating valence and conduction bands.

(b) Silicon $E_g = 1.1\,eV$. Let $m_h^* = m_e^* = \frac{1}{2} m_e$.

Use

$$n(T_{room}) \approx \frac{1}{\sqrt{2}} \left(\frac{1}{2}\right)^{3/2} \pi^{-3/2} \left(k_B T \frac{m_e}{\hbar^2}\right)^{3/2} e^{-\frac{E_g}{2k_B T}}$$

Do calculation in Mathematica (upload "PhysicalConstants package)

Then $e^{-E_g/2k_BT} \longrightarrow 5.77 \cdot 10^{-10}$

while $\frac{1}{4}\left(\frac{k_B T\, m_e}{\pi \hbar^2}\right)^{3/2} = 8.87 \cdot 10^{24}\, m^{-3}$ (remember that $eV = 1.602 \cdot 10^{-19}\, J$, $J = kg \cdot m^2/s^2$)

at $T = 300K$.

$\Rightarrow n_{Si} = 5.12 \cdot 10^{15}\, m^{-3}$.

For Germanium $E_g = 0.75\,eV \Rightarrow e^{-\frac{E_g}{2k_BT}} = 5.02 \cdot 10^{-7}$

so that $n_{Ge} = 4.46 \cdot 10^{18}\, m^{-3}$.

(My previous calculation had numerical mistakes)

"Intrinsic" behavior will persist for as long as

$n_{donors} < n_{intrinsic} \approx 10^{16}\, m^{-3}$.

(c) Fig. 17.6 shows that at low T $n \sim 2 \cdot 10^{19}\, m^{-3}$.

The slope is $\approx \frac{\log(10)}{0.002} = 500 \Rightarrow \log(n) \sim \frac{-500}{T}$

$$n = 10^{-\left(\frac{500}{T}\right)} = e^{-\frac{500}{T}\ln(10)} \simeq e^{-1150/T}$$

But $n \sim e^{-E_g/2k_BT} \Rightarrow E_g \approx 2\, k_B \cdot 1150\, K = 8.6 \cdot 10^{-5} \frac{eV}{K} \cdot 2300$

$\approx 0.2\, eV$.

Prob. 17.3

From the previous problem $\quad 1 = \frac{n(T)}{p(T)} = \left(\frac{m_e}{m_h}\right)^{3/2} \frac{e^{-\beta(\varepsilon_c-\mu)}}{e^{-\beta(\mu-\varepsilon_v)}}$

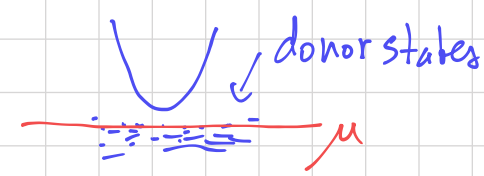
$$\Rightarrow \quad e^{\beta(2\mu-\varepsilon_c-\varepsilon_v)} = \left(\frac{m_h}{m_e}\right)^{3/2}$$

$$\Rightarrow \quad \mu = \frac{1}{2}(\varepsilon_c+\varepsilon_v) + \frac{3}{4}k_B T \ln\left(\frac{m_h}{m_e}\right)$$

(b) Semiconductor doped with donors:

$n = n_{intrinsic} + n_{donors} \simeq n_{donors}$

as typically donors density is much greater than that of intrinsic electron one



$$\Rightarrow \quad n_{donors} = \frac{1}{4}\left(\frac{2m_e k_B T}{\pi\hbar^2}\right)^{3/2} e^{-\beta(\varepsilon_c-\mu)}$$

and solve for μ: $\quad \beta(\mu-\varepsilon_c) = \ln\left[\frac{n_{donors}}{\frac{1}{4}\left(\frac{2m_e k_B T}{\pi\hbar^2}\right)^{3/2}}\right]$

$$\mu = \varepsilon_c + k_B T \ln[\ ]$$

so that μ is near the conduction band edge, ε_c (but below it) and moves down towards the middle of the gap as T increases: $\mu = \varepsilon_c - k_B T \ln\left(\frac{k_B T}{\dots}\right)$

For acceptor doping the situation is opposite: here $p = p_{acceptor}$

$$\Rightarrow \quad \mu \simeq \varepsilon_v - k_B T \ln\left[\frac{p_{acceptor}}{\frac{1}{4}\left(\frac{2m_h k_B T}{\pi\hbar^2}\right)^{3/2}}\right] \to \varepsilon_v + T\ln T$$

Chem. potential is near ε_v at low T and moves up with T.